

Real Numbers

Class 10 Mathematics (NCERT) - Notes Solutions

1. The Fundamental Theorem of Arithmetic

The Fundamental Theorem of Arithmetic is a key concept regarding the nature of composite numbers. It states that every composite number can be expressed (factorised) as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur [1].

For example, the number 32760 can be factorised as $2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 \times 13$, or written with powers as $2^3 \times 3^2 \times 5 \times 7 \times 13$. Once we decide to write the primes in ascending order, the factorisation is unique [2].

Applications:

This theorem is primarily used to find the HCF (Highest Common Factor) and LCM (Lowest Common Multiple) of positive integers [3].

Relationship between HCF and LCM:

For any two positive integers a and b , the following relationship holds true:

$$HCF(a, b) \times LCM(a, b) = a \times b$$

Note: This formula applies only to **two** numbers. It does not hold for three or more numbers [4, 5].

- HCF = Product of the smallest power of each common prime factor [4]
- LCM = Product of the greatest power of each prime factor involved [4]
- Formula: $HCF(a, b) \times LCM(a, b) = a \times b$

2. Revisiting Irrational Numbers

An irrational number is a number that cannot be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$. Examples include $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, π , etc [6].

Important Theorem (Theorem 1.2):

Let p be a prime number. If p divides a^2 , then p divides a , where a is a positive integer [7]. This theorem is essential for proving the irrationality of numbers like $\sqrt{2}$.

Proof Method:

We use the method of **Proof by Contradiction**. We assume the number is rational, show that this assumption leads to a mathematical contradiction (usually involving common factors), and thus conclude the number must be irrational [8].

- If prime p divides a^2 , then p divides a [7]
- $\sqrt{2}, \sqrt{3}, \sqrt{5}$ are proved irrational using contradiction [6]
- Sum/difference of rational and irrational is irrational [9]

Exercise 1.1: Detailed Solutions

Q1. Express each number as a product of its prime factors:

(i) 140



Steps:

Divide by the smallest prime factor repeatedly.

$$140 = 2 \times 70$$

$$70 = 2 \times 35$$

$$35 = 5 \times 7$$

Result: $140 = 2^2 \times 5 \times 7$ [5]

(ii) 156

$$156 = 2 \times 78$$

$$78 = 2 \times 39$$

$$39 = 3 \times 13$$

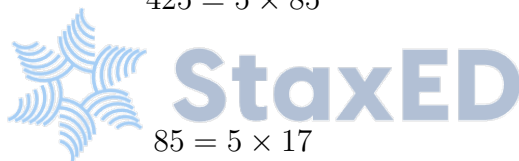
Result: $156 = 2^2 \times 3 \times 13$ [5]

(iii) **3825**

$$3825 = 3 \times 1275$$

$$1275 = 3 \times 425$$

$$425 = 5 \times 85$$



$$85 = 5 \times 17$$

Result: $3825 = 3^2 \times 5^2 \times 17$ [5]

(iv) **5005**

$$5005 = 5 \times 1001$$

$$1001 = 7 \times 143$$

$$143 = 11 \times 13$$

Result: $5005 = 5 \times 7 \times 11 \times 13$ [5]

(v) **7429**

$$7429 = 17 \times 437$$

$$437 = 19 \times 23$$

Result: $7429 = 17 \times 19 \times 23$ [5]

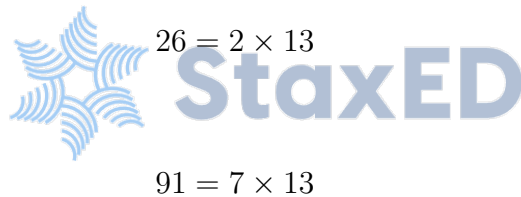
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Q2. Find the LCM and HCF of the following pairs of integers and verify that $\text{LCM} \times \text{HCF} = \text{product of the two numbers}$.

(i) 26 and 91

Given: Numbers 26 and 91.

Prime Factorisation:



$$26 = 2 \times 13$$

$$91 = 7 \times 13$$

HCF: Product of smallest power of common primes = 13

LCM: Product of greatest power of all primes = $2 \times 7 \times 13 = 182$

Verification:

$$\text{LCM} \times \text{HCF} = 182 \times 13 = 2366$$

$$\text{Product} = 26 \times 91 = 2366$$

Conclusion: LHS = RHS. Verified [10].

(ii) 510 and 92

Prime Factorisation:

$$510 = 2 \times 3 \times 5 \times 17$$

$$92 = 2^2 \times 23$$

HCF: Common factor is 2. $HCF = 2$

LCM: $2^2 \times 3 \times 5 \times 17 \times 23 = 23460$

Verification:

$$LCM \times HCF = 23460 \times 2 = 46920$$

$$Product = 510 \times 92 = 46920$$

Conclusion: LHS = RHS. Verified [10].

(iii) **336 and 54**

Prime Factorisation:



$$336 = 2^4 \times 3 \times 7$$

$$54 = 2 \times 3^3$$

HCF: $2^1 \times 3^1 = 6$

LCM: $2^4 \times 3^3 \times 7 = 3024$

Verification:

$$LCM \times HCF = 3024 \times 6 = 18144$$

$$Product = 336 \times 54 = 18144$$

Conclusion: LHS = RHS. Verified [10].

Q3. Find the LCM and HCF of the following integers by applying the prime factorisation method.

(i) 12, 15 and 21

$$12 = 2^2 \times 3$$

$$15 = 3 \times 5$$

$$21 = 3 \times 7$$

HCF: Common prime is 3. HCF = 3

LCM: $2^2 \times 3 \times 5 \times 7 = 420$ [10]

(ii) 17, 23 and 29

$$17 = 17 \times 1$$

$$23 = 23 \times 1$$

$$29 = 29 \times 1$$

(All are prime numbers)

HCF: 1

LCM: $17 \times 23 \times 29 = 11339$ [10]

(iii) 8, 9 and 25

$$8 = 2^3$$

$$9 = 3^2$$

$$25 = 5^2$$

(No common prime factors)

HCF: 1

LCM: $2^3 \times 3^2 \times 5^2 = 8 \times 9 \times 25 = 1800$ [10]

Q4. Given that $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$.

Given: $a = 306, b = 657, \text{HCF} = 9$.

To Find: LCM.



Method: Use formula $\text{LCM} \times \text{HCF} = a \times b$.

$$\text{LCM} = \frac{a \times b}{\text{HCF}}$$

$$\text{LCM} = \frac{306 \times 657}{9}$$

$$\text{LCM} = 34 \times 657$$

Result: $\text{LCM} = 22338$ [10].

Q5. Check whether 6^n can end with the digit 0 for any natural number n .

Reasoning:

If a number ends with the digit 0, it must be divisible by 10, meaning it must be divisible by both 2 and 5.

The prime factorisation of 6^n is:

$$6^n = (2 \times 3)^n = 2^n \times 3^n$$

The prime factors of 6^n are only 2 and 3.

By the uniqueness of the Fundamental Theorem of Arithmetic, there are no other primes in the factorisation of 6^n . Since the prime factor 5 is missing, 6^n can never end with the digit 0 [10, 11].

Result: No, 6^n cannot end with digit 0.

Q6. Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers.

Reasoning: A composite number has factors other than 1 and itself.

Case 1: $N_1 = 7 \times 11 \times 13 + 13$

Take 13 as a common factor:

$$N_1 = 13 \times (7 \times 11 + 1)$$

$$N_1 = 13 \times (77 + 1) = 13 \times 78$$

Since N_1 has factors 13 and 78 (other than 1), it is composite.

Case 2: $N_2 = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

Take 5 as a common factor:

$$N_2 = 5 \times (7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1)$$

$$N_2 = 5 \times (1008 + 1) = 5 \times 1009$$

Since N_2 has factors 5 and 1009, it is composite [10].

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Q7. Circular path problem: Sonia takes 18 mins, Ravi takes 12 mins. When will they meet again at the starting point?

To Find: The smallest time interval divisible by both 18 and 12 (LCM).

Prime Factorisation:

$$18 = 2 \times 3^2$$

$$12 = 2^2 \times 3$$

Calculation:



$$LCM(18, 12) = 2^2 \times 3^2 = 4 \times 9 = 36$$

Result: They will meet again after **36 minutes** [10].

Exercise 1.2: Detailed Solutions

Q1. Prove that $\sqrt{5}$ is irrational.

Proof:

Let us assume, to the contrary, that $\sqrt{5}$ is rational.

Therefore, we can write $\sqrt{5} = \frac{a}{b}$, where a and b are coprime integers and $b \neq 0$.

$$\sqrt{5} = \frac{a}{b} \implies \sqrt{5}b = a$$

Squaring both sides:

$$5b^2 = a^2 \quad \dots (1)$$

This means a^2 is divisible by 5. By Theorem 1.2, **a is also divisible by 5.**

So, we can write $a = 5c$ for some integer c .

Substituting $a = 5c$ into equation (1):

$$5b^2 = (5c)^2$$

$$5b^2 = 25c^2$$

$$b^2 = 5c^2$$

This means b^2 is divisible by 5. Therefore, **b is also divisible by 5.**

Contradiction:



We have shown that both a and b have at least 5 as a common factor. But this contradicts the fact that a and b are coprime. This contradiction has arisen because of our incorrect assumption that $\sqrt{5}$ is rational.

Conclusion: $\sqrt{5}$ is irrational [12].

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Q2. Prove that $3 + 2\sqrt{5}$ is irrational.

Proof:

Let us assume, to the contrary, that $3 + 2\sqrt{5}$ is rational.

Then we can find coprime integers a and b ($b \neq 0$) such that:

$$3 + 2\sqrt{5} = \frac{a}{b}$$

Rearranging to isolate $\sqrt{5}$:

$$2\sqrt{5} = \frac{a}{b} - 3$$

$$2\sqrt{5} = \frac{a - 3b}{b}$$

$$\sqrt{5} = \frac{a - 3b}{2b}$$

Since a and b are integers, $\frac{a-3b}{2b}$ is rational.

Therefore, $\sqrt{5}$ is rational.

Contradiction:

But we know that $\sqrt{5}$ is irrational. This contradiction implies our assumption was false.

Conclusion: $3 + 2\sqrt{5}$ is irrational [12].

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Q3. Prove that the following are irrationals:

(i) $\frac{1}{\sqrt{2}}$

Proof:

Assume $\frac{1}{\sqrt{2}}$ is rational.

$$\frac{1}{\sqrt{2}} = \frac{a}{b} \quad (a, b \text{ are coprime integers, } b \neq 0)$$

$$b = a\sqrt{2}$$

$$\sqrt{2} = \frac{b}{a}$$

Since a, b are integers, $\frac{b}{a}$ is rational, which implies $\sqrt{2}$ is rational. This contradicts the fact that $\sqrt{2}$ is irrational.

Conclusion: $\frac{1}{\sqrt{2}}$ is irrational [12].

(ii) $7\sqrt{5}$

Proof:

Assume $7\sqrt{5}$ is rational.

$$7\sqrt{5} = \frac{a}{b}$$

$$\sqrt{5} = \frac{a}{7b}$$

Since a, b are integers, $\frac{a}{7b}$ is rational. This implies $\sqrt{5}$ is rational, which contradicts the fact that $\sqrt{5}$ is irrational.

Conclusion: $7\sqrt{5}$ is irrational [12].

(iii) $6 + \sqrt{2}$

Proof:

Assume $6 + \sqrt{2}$ is rational.



$$6 + \sqrt{2} = \frac{a}{b}$$

$$\sqrt{2} = \frac{a}{b} - 6$$

$$\sqrt{2} = \frac{a - 6b}{b}$$

Since a, b are integers, $\frac{a-6b}{b}$ is rational. This implies $\sqrt{2}$ is rational, which contradicts the fact that $\sqrt{2}$ is irrational.

Conclusion: $6 + \sqrt{2}$ is irrational [12].